

Replication Report: Faraday and Kerr Rotation due to Photoinduced Orbital Magnetization in a 2DEG

Durnev (2023), arXiv:2306.08509

Independent analytic replication

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Abstract

We independently replicate the central analytic results of Durnev (2023), which develops a Boltzmann/kinetic-theory of the pump-induced Faraday and Kerr rotation of a probe field in a two-dimensional electron gas (2DEG). A circularly polarized pump induces an orbital magnetization (intraband inverse Faraday effect), producing a transverse conductivity σ_{xy} and hence rotation of a linearly polarized probe. Working in Gaussian/CGS units, we implement the paper's closed-form expressions for the graphene (linear-dispersion) case and evaluate them at the resonance condition $\Omega\tau_1 \approx 1$, $\omega \approx \Omega$. All intermediate cross-checks reproduce the paper's own stated numbers to $< 1\text{--}2\%$: the Fermi energy $\varepsilon_F = 63.9\text{ meV}$ (paper 64 meV), the dimensionless coupling $2\pi\sigma_0/(c\bar{n}) = 0.0708$ (paper 0.071), and the transmission $T(\Omega) = 0.699$ (paper range 0.63–0.70). Two independent formulas for the peak Faraday angle agree ($\theta_F = 0.0441^\circ$ from Eq. 26 and 0.0417° from Eq. 5+25), and the synthetic magnetic field $B_{\text{syn}} = 0.088\text{ T}$ agrees with the paper's $\sim 0.1\text{ T}$ estimate to 12%. **Verdict: REPLICATED** (coverage 7/10, agreement 9/10).

1 Paper and Claim

Durnev (2023) studies the Faraday and Kerr rotation of a weak probe field arising from the orbital magnetization of a 2DEG driven by a circularly polarized pump, in the intraband spectral range. The theory is built on the analytic solution of the kinetic (Boltzmann) equation for both linear (graphene) and parabolic dispersions and arbitrary scattering potential. The headline claim: at the resonance $\omega \approx \Omega$ with $\Omega\tau_1 \sim 1$, the Faraday and Kerr rotation angles reach $\sim 0.1^\circ$ per 1 kW/cm^2 of pump intensity in graphene, corresponding to a pump-induced synthetic magnetic field of $\sim 0.1\text{ T}$.

2 Method

This is an analytic/closed-form paper: the replication *is* the derivation and its numerical evaluation. We reimplement, from the paper's equations, the graphene single-layer (linear-dispersion) chain in Gaussian/CGS units:

- **Eq. (5)** $\theta_F + i\epsilon_F = 2\pi\sigma_{xy}/[c\bar{n}(1+\alpha)]$, with $\alpha = 2\pi\sigma/(c\bar{n})$ and $\bar{n} = (n_1 + n_2)/2$.
- **Eq. (25)** the resonance transverse conductivity $\sigma_{xy}(\omega)$ for a linear spectrum.
- **Eq. (26)** the explicit closed-form resonance Faraday angle $\theta_F(\omega)$.
- **Eq. (27)** the synthetic magnetic field $B_{\text{syn}} \sim ec|E_\Omega|^2\tau_0/\varepsilon_F$.

- Field at the sheet: $|E_\Omega|^2 = 2\pi T(\Omega)I_\Omega/(cn_2)$, with $T(\Omega) = n_2|\bar{t}|^2/n_1$ and $\bar{t} = t_{12}/(1 + \alpha(\Omega))$.

Supporting relations: graphene Fermi energy $\varepsilon_F = \hbar v_0 \sqrt{\pi n_e}$ and static conductivity $\sigma_0 = e^2 v_0^2 n_e \tau_1 / \varepsilon_F$. Numerical evaluation uses NumPy; the probe frequency ω is scanned across the resonance to locate the peak angle.

3 Parameters

Symbol	Value	Meaning
τ_1	0.1 ps	momentum relaxation time ($\Omega\tau_1 = 1$)
τ_0	5 ps	energy / zeroth-harmonic relaxation time
v_0	1×10^8 cm/s	graphene Fermi velocity
n_e	3×10^{11} cm $^{-2}$	2D electron density
I_{pump}	1 kW/cm 2	pump intensity
n_1, n_2	1, 3	refractive indices (substrate case)
P_{circ}	1	pump circular polarization degree

Table 1: Graphene-on-substrate parameters (Fig. 3 of the paper).

4 Results and Cross-Checks

Quantity	This work	Paper	Agreement
ε_F (meV)	63.90	64.0	< 0.2%
$2\pi\sigma_0/(c\bar{n})$	0.0708	0.071	< 1%
$T(\Omega)$	0.699	0.63–0.70	in range
Ω_{res} (THz)	1.59	($\Omega\tau_1 = 1$)	consistent
θ_F peak, Eq. 26 (deg)	0.0441	~ 0.05 (Fig.3)	$\approx$
θ_F peak, Eq. 5+25 (deg)	0.0417	~ 0.05 (Fig.3)	$\approx$
B_{syn} (T)	0.088	~ 0.1	12%

Table 2: Cross-checks and headline results. The two independent Faraday formulas (Eq. 26 and Eq. 5+25) agree with each other to $\sim 6\%$.

The intermediate cross-checks are the strongest evidence: they reproduce the paper’s *own* intermediate numbers to < 1–2%, confirming the unit system, the Fermi-energy/conductivity relations, and the transmission model are all faithfully reproduced. The peak Faraday angle $\theta_F \approx 0.044^\circ$ lands squarely on the $\Omega\tau_1 = 1$ graphene-on-substrate curve of Fig. 3 (y-axis $\sim 0.05^\circ$). The headline “0.1 $^\circ$ ” is an order-of-magnitude statement spanning Figs. 2–4 (0.1–1 $^\circ$); the graphene-on-substrate value sits correctly within it. The synthetic field $B_{\text{syn}} = 0.088$ T (paper’s explicit estimate parameters $\varepsilon_F = 50$ meV, $\tau_0 = 10$ ps) matches ~ 0.1 T to 12%.

5 Verdict

REPLICATED. Coverage $\approx 7/10$, Agreement $\approx 9/10$. For an analytic/closed-form paper the replication is the re-derivation and numerical evaluation of the governing formulas. Multiple *independent* cross-checks match the paper to $< 1\text{--}2\%$, and two independent Faraday-angle formulas mutually agree to $\sim 6\%$. Honest gaps: analytic-only (no full transport simulation); only the linear/graphene branch is coded (parabolic / bilayer Eqs. 21/23/24 not implemented); Kerr angle taken as $\theta_K \approx -\theta_F$ in the large-contrast limit rather than evaluated independently via Eq. (6); and no pixel-level digitization of the paper’s figures was performed.